Ericsson PE Node JAY-PE-09 Jayanagar Metro Site QoS Congestion Event SOP

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_4441

Description

The failure domain of QoS congestion events on Ericsson node JAY-PE-09, specifically involving microbursts, interface tail drops, and QoS queue starvation, is primarily attributed to the node's role as a Packet Engine (PE) in the Service Aware Networking (SAN) architecture. The PE node processes and forwards traffic on behalf of the Service Router (SR), utilizing the node's high-speed interface and packet processing capabilities. However, the PE node's reliance on the Shared Resource Module (SRM) for QoS configuration and policing may lead to congestion events when the SRM fails to properly manage traffic. Furthermore, the PE node's use of the Traffic Management System (TMS) for traffic policing and shaping may also contribute to congestion events when the TMS fails to adapt to changing traffic conditions. In addition, the PE node's interface configuration, including the use of traffic shaping and policing mechanisms, may also influence the likelihood of congestion events. To mitigate these risks, regular monitoring and maintenance of the PE node's QoS configuration and policing mechanisms, as well as the SRM and TMS, are essential to prevent congestion events and ensure optimal network performance.

Pre-requisite Checks

- Dispatch remote hands to Jayanagar Metro Site for physical site visit and equipment access

- Verify the redundant link status between PE node JAY-PE-09 and the adjacent nodes

- Check the power supply and cooling systems at the site to ensure they are functioning within normal parameters

Potential Root Causes

Cause	Probability
Shared Resource Module (SRM) failure to properly manage QoS traffic	High
Traffic Management System (TMS) failure to adapt to changing traffic conditions	Medium
Interface configuration issues, including traffic shaping and policing mechanisms	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policing, Expected Output: QoS policing configuration and statistics, Description: This command displays the QoS policing configuration and statistics for the PE node, including the policing thresholds, action profiles, and drop rates. This information can be used to identify potential congestion events and isolate the root cause of the issue.
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Command: show interface stats brief, Expected Output: Interface statistics, including errors and drop rates, Description: This command displays a brief summary of the interface statistics for the PE node, including errors, drops, and packet counts. This information can be used to identify potential interface-related congestion events and isolate the root cause of the issue.

Command: show tms status, Expected Output: TMS status and configuration, Description: This command displays the TMS status and configuration for the PE node, including the TMS version, configuration, and operational status. This information can be used to identify potential TMS-related congestion events and isolate the root cause of the issue.
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Resolution Steps

Step 1: Step: Reboot the Shared Resource Module (SRM), Command: reboot srmodule, Description: Rebooting the SRM may resolve congestion events caused by SRM failures to properly manage QoS traffic.

Step 2: Step: Reconfigure the Traffic Management System (TMS), Command: config tms action-profile , Description: Reconfiguring the TMS may resolve congestion events caused by TMS failures to adapt to changing traffic conditions.

Step 3: Step: Reconfigure the interface, Command: config interface policing , Description: Reconfiguring the interface may resolve congestion events caused by interface configuration issues, including traffic shaping and policing mechanisms.

Post-Check Verification

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Command: show qos policing, Expected Output: QoS policing configuration and statistics, Description: Verify that the QoS policing configuration and statistics have been updated correctly.

Command: show interface stats brief, Expected Output: Interface statistics, including errors and drop rates, Description: Verify that the interface statistics have been updated correctly.

Command: show tms status, Expected Output: TMS status and configuration, Description: Verify that the TMS status and configuration have been updated correctly.
```

Roll-back Procedures

- Step: Reboot the Shared Resource Module (SRM), Command: reboot srmodule, Description: If the fix fails, rebooting the SRM may restore the previous configuration and resolve the issue.

- Step: Reconfigure the Traffic Management System (TMS), Command: config tms action-profile , Description: If the fix fails, reconfiguring the TMS may restore the previous configuration and resolve the issue.

- Step: Reconfigure the interface, Command: config interface policing , Description: If the fix fails, reconfiguring the interface may restore the previous configuration and resolve the issue.